

Advanced Java Programming

CSIT — 7th Semester — 2080 — Answer Sheet

Subject Code: CSC409

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Why do we need to synchronize the thread? Justify with an example. An array with an odd number of elements is said to be centered if all elements (except the middle one) are strictly greater than the value of the middle element. Note that only arrays with an odd number of elements have a middle element. Write a function that accepts an integer array and returns 1 if it is a centered array, otherwise it returns 0?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. How does JSP differ with Servlet and show the life cycle of Servlet? How do you create and read the cookies and session using JSP? Illustrate with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Describe any two types of layout manager. Using swing components design a form with three buttons with captions "RED", "BLUE" and "GREEN" respectively. Then write a program to handle the event such that when the user click the button, the color of that button will be the same as its caption.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. List the steps to create RMI application. Differentiate between RMI and CORBA.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Write a JavaFX application with components, buttons, textfields and labels, arranged in a VBox or HBox layout.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Assume a table MOVIE(id, title, genre). Now using JDBC, perform the following queries.
a. Add any three records to the MOVIE table.
b. Using prepared statement, update the genre to "Comedy" having title "Jatra"

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. When do we need internal frame? How do you create table using swing?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What do you mean by JSP implicit objects? Discuss about Java Mail API.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Describe the role of Result Sets. What is wrong in the following code?
`public class Point{int p;public void setP(int p){p = p;}}`

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Write a TCP client - server system in which the client program sends two integers to a server program which returns the greatest among them.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Suppose that 9 integers are written in a file named "magic.txt" in the arrangement of 3 x 3 separated by space in. Write a program, whether the integers in all rows, all columns and both diagonals sum to the same constant or not.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write a program to create a menu named "File" with menu items "New", "Save" and "Exit".

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.